A replication and formalization of the three-trait QTL SEM in Mi (2009) and Mi, Eskridge, Wang, and coauthors (2010)

27 October 2026

This manuscript was drafted by a language model as a replication and formalization record. The date is an internal manuscript date. The note is not a journal article, and it does not speak for the authors of the sources it studies.

Abstract

This note records an attempt by a language model to replicate, and to formalize in Lean, the three-trait QTL structural equation model associated with Mi's 2009 University of Nebraska–Lincoln dissertation, with Mi, Eskridge, Wang, Baenziger, Campbell, Gill, Dweikat, and Bovaird, Statistical Applications in Genetics and Molecular Biology 9 (2010), article 38, and with the 2008 conference version of that design. The dissertation was not obtained. On the pattern printed in equations (7)–(8) of the 2010 paper, diagonal residuals are a left inverse of the covariance, each single residual covariance collides, and each of the three bow-free patterns is a left inverse. Those statements are Lean proofs, with no `sorry`. A first reading of equation (7), with one minus in front of M, does not reproduce Table 2's total effect 0.75. Measuring the glyph positions puts the minuses on individual entries. That reading reproduces the parameter block, including 0.75, and raises the single-trait power for the third QTL on the first trait from 13% to 86% in 1000 replicates. The printed multi-trait powers 64, 77, and 96 are not jointly reproduced by any design this note could reconcile with the simulation paragraph. The closest percentage triple found is 65.2, 73.3, 99.3. A product of binomial probabilities for 64, 77, and 96 at the baseline rates is about 1.4×10^{-12}, under an independence assumption the three detections do not satisfy. Two further comparisons are left as source discrepancies under our reading: Table 2 prints -0.435 and 0.065 where the arithmetic gives -0.4375 and 0.0625, and Table 1 prints SEM powers above the least-squares powers on a model where the two likelihood ratios coincide.

Question, source, and answer

Question	Can the trait residuals in the Mi et al. (2010) QTL SEM be correlated after environment and block effects are removed? Printed page 18 (PDF page 20).
Source	Mi et al., SAGMB 9 (2010), article 38. The request is not a numbered conjecture.
Answer on the printed pattern	On equation (7), diagonal residuals are a left inverse of the covariance. Each single residual covariance collides. Each bow-free missing-edge pattern is a left inverse, and the plug-in is almost surely consistent. Theorems 246, 114, 187, and 270.
What the simulations add	A seeded replication, the bias and size distortion of the diagonal fit, and coverage of `recoverBow`. These are sampled experiments, not Lean proofs.
Still open	A fit to the wheat measurements, the general mixed-graph characterization, and a Lean central limit theorem.

The manuscript date 27 October 2026 is an internal date. It is not an external publication date. Elapsed times below are measured to that date. A year-only source is reported as a range.

Source	Earliest date used	Inspected version	Elapsed to 27 Oct 2026
Brito and Pearl, bow-free identification	2002 (year)	SEM 9, 459–474	23–24 years
Campbell et al., chromosome 3A QTLs	2003 (year)	Crop Sci. 43, 1493–1505	22–23 years
Li et al., structural QTL model	21 July 2006	PLoS Genet. 2, e114	20 years, 3 months, 6 days
Mi et al., the source question	2010 (year)	SAGMB 9, article 38	15 years 10 months to 16 years 10 months
Valente et al., recursive search	June 2010 (month)	Genetics 185, 633–644	16 years, 4 months (month precision)
Drton, Foygel, and Sullivant	arXiv 4 Mar 2010; online 13 May 2011	Ann. Statist. 39, 865–886	15 years, 5 months, 14 days from the journal date
Foygel, Draisma, and Drton, half-trek	arXiv 27 July 2011; online 3 Oct 2012	Ann. Statist. 40, 1682–1713	14 years, 24 days from the journal date
Hidalgo-Contreras et al.	28 September 2021	Agronomy 11, 1953	5 years, 29 days
This manuscript	27 October 2026	internal manuscript date	—

1 What this replication did

The work in this directory is a formalization of one identification question and a set of seeded simulations aimed at Table 1 and Table 2 of Mi et al. (2010). The sections after this one keep the working record: the quoted request, the glyph coordinates, the Lean statements, the design grid, and the bow-free and Campbell calculations. Commands and file paths are collected in Appendix B.

1.1 The Lean formalization

The Lean development is `formalization/correlated-residuals/CorrelatedResiduals/`, project file `lakefile.toml`, toolchain Lean 4.22.0 and Mathlib v4.22.0. `lake build` is the build. The proofs use no `sorry` and no `sorryAx`. `#print axioms` on the theorems named below reports `propext`, `Classical.choice`, and `Quot.sound`. `Classical.choice` enters through the matrix inverse and the positive-definite unit argument. `Quot.sound` is the construction of $\mathbb{R}$. *(our own Lean proof)*

The statements, with file and line, are:

- `recover_sigma` (`Diagonal.lean`:246). On the equation-(7) zero pattern, positive diagonal Ψ and positive definite Φ make (B, Γ, Ψ, Φ) a function of Σ.
- `bow12_same_law` (`Collision.lean`:114), and the two sibling collisions for the other pairs. Each one-edge residual extension collides.
- `recover_bow` (`BowFree.lean`:187), `recover_bow01` (`BowMissing.lean`:148), and `recover_bow23` (`BowMissing.lean`:353). Each bow-free missing-edge pattern is a left inverse.

- `unitUpper_factor_unique` (`Triangular.lean`:68) and `sigmaRec_injective` (`Triangular.lean`:298).
- `plug_in_consistent_ae` (`Law.lean`:174) and `plug_in_bow_consistent_ae` (`BowConsistency.lean`:270), with the two further bow patterns. Almost-sure consistency of the plug-in. This is not a Lean central limit theorem.

The Gaussian delta method is a written derivative and a numeric check (`delta_method.py`), not a Lean proof. The listings in Appendix A are the theorem bodies at commit `05d69ea141adf3d5ee936d3b48061bde241250dc`.

1.2 The simulation studies

Three seeded studies live in `simulation/studies.py`. Study 1 is a 100-replicate scan under the earlier reading of equation (7). Study 2 is the bow-free Monte Carlo, 400 replicates, seed `20101028`: the diagonal plug-in sends the missing-edge coefficient to 0.8ρ, and `recoverBow` interval coverage at the nominal 95% level falls between about 0.915 and 0.958. Study 3 refits B and Ψ to three phenotypic correlations from Campbell et al. (2003) and reruns the scan. That run is a scale check. It is not a replication of Table 1, and the Campbell PDF itself was not stored. *(our own simulation)*

A separate grid, `simulation/gaps.py`, reruns the scan under one change at a time, including 1000-replicate versions of the earlier reading and the glyph reading. `simulation/assumptions.py` crosses the choices the simulation paragraph does not pin down, at 1000 replicates, and also writes a noncentral chi-square approximation. `qtl1_search.py` moves the first QTL, the sample size, the marker gap, and the cofactor rule, again at 1000 replicates, and records whether each choice is consistent with the printed sentence.

1.3 The replication results

Under the glyph reading, $n = 150$, seed `20101101`, and 1000 replicates, the wide-window SEM powers are 50.8, 53.5, and 86.1. A second seed, `20101102`, gives 49.4, 48.3, and 84.6. Single-trait powers for Y_1 are 3.2, 1.2, and 88.6, against the printed 6, 3, and 97. Table 1 prints SEM powers 64, 77, and 96, least-squares powers 57, 69, and 94, single-trait powers 6/3/97, 29/43/33, and 10/1/17, and overall powers 40, 45, and 97. *(our own simulation, set next to a cited table)*

The glyph total-effect matrix is

$$(I - M)^{-1}G = \begin{pmatrix} 1/8 & 0 & 3/4 \\ 3/8 & -7/16 & 3/8 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

The $(1, 3)$ entry is the 0.75 in Table 2. The same arithmetic gives $-7/16 = -0.4375$ where Table 2 prints -0.435, and the indirect piece $1/16 = 0.0625$ where the table prints 0.065. Those two cells are discussed with the unresolved comparisons. The 1000-replicate single-trait power for QTL3 on Y_1 is 86.2% (seed `20101041`). The earlier reading, which produced the total effect 0.25, gave 13% for that cell.

No configuration that stays with the printed positions, the printed sample size, and the printed threshold reproduces 64, 77, and 96 together. The smallest squared error among the percentage triples that were computed is 65.2, 73.3, 99.3, from QTL1 at 24 cM and $n = 250$. That design is not the design in the simulation sentence. Section 4 records what was tried and what is still open.

2 Assumptions

Each subsection is one assumption used to turn the printed paragraph into a program. A choice the paragraph states is not listed here as an assumption. The grid that turns one assumption off at a time is Section 12.

2.1 Design

The population is an F1 selfed for twelve generations, with residual heterozygotes coded ± 1 at equal probability. Mi et al. (2010) say recombinant inbred lines and do not give the generation count or a segregation-distortion model. The map function in the SAGMB scan is Haldane. The 2008 conference note names Haldane and prints the infinite-RIL expansion $2r/(1+2r)$; the SAGMB simulation paragraph does not. Cofactors are every marker except the flanking pair, which is the sentence on PDF page 10. The conditional expectation of the QTL genotype given the flanking markers follows Haley and Knott (1992), as the methods section says, implemented as the exact two-point expectation on the twelve-generation pedigree. *(assumption, flagged)*

2.2 Map positions

Markers are placed at $0, 10, \ldots, 100$ cM. The sentence says eleven evenly spaced markers on a 100 cM chromosome. Putting a marker at each end is a reading of "evenly spaced," not a coordinate the paper prints. QTLs are placed at 22, 42, and 78 cM, which the sentence does print. The scan step is 1 cM. A detection is counted in a window of ± 10 cM around the true position, and the position standard deviations use that window. The paper's position standard deviations are consistent with a window nearer ± 5 cM, which the text does not state. The ± 10 cM window is ours. *(assumption, flagged)*

2.3 Sample size

The SAGMB simulation uses 150 lines and 100 replicates. The baseline powers quoted above use 1000 replicates of that 150-line design, so that Monte Carlo error is smaller than the gap under discussion. The 100-replicate size is what Table 1 itself is. Companion designs that were simulated and then set aside are 250 lines (Mi et al., Genetics Research, 2010) and 300 lines (the 2008 conference note). Neither is the SAGMB sample size. *(assumption, flagged where the replicate count exceeds the printed* 100*)*

2.4 Thresholds

A multi-trait detection uses $\chi^2_{0.005,4} = 14.86$. A single-trait detection uses $\chi^2_{0.005,2} = 10.60$. Both numbers are printed. Treating 14.86 as a LOD score, or as a χ^2 critical value on 3 or on 1 degree of freedom, was tested and is not an assumption of the baseline. The permutation threshold, a 95% quantile near 16.1 under the baseline population, was computed for comparison and is not the threshold in the scan that produced 50.8, 53.5, and 86.1. *(the printed thresholds, used as printed)*

2.5 Readings of equation (7)

Two readings were programmed.

The earlier reading puts the displayed minus in front of M and takes every entry of G as positive, so $B = -M$. The total effect of QTL3 on Y_1 is then $1/4$, and the single-trait power for that cell is

13% at 100 replicates. This reading is study 1 of `studies.py` and the rows of `gaps.py` labelled as the previous reading.

The glyph reading is $y = My + Gx + \varepsilon$, with

$$M = \begin{pmatrix} 0 & 1/2 & 1/4 \\ 0 & 0 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1/8 & 1/4 & 1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/4 & -1/8 & 1/4 \end{pmatrix},$$

so $B = M$, and $\Psi = \mathrm{diag}(1.6, 1.8, 2.5)$ as printed in equation (8). The baseline, the joint search, and the Table 2 arithmetic use this reading. How the signs were assigned is the next section. *(our reading of the glyph positions)*

3 Ambiguities that were resolved

3.1 The sign in front of M

The first programs read equation (7) as $y = -My + Gx + \varepsilon$. That population does not produce Table 2's total effect 0.75, and it produces a single-trait power of 13% where Table 1 prints 97% for QTL3 on Y_1.

The PDF content stream on page 10 of the ARS copy of Mi et al. (2010) settles the operator. The equals sign ends at $x = 145.9$ points. The parenthesis of M starts at $x = 148.1$. No minus sits between them. Four unary minuses are present, and each ends on the first digit of the number it signs: $G_{11} = -0.125$ (the minus ends at $x = 250.7$), $M_{23} = -0.5$ (ends at $x = 194.5$), $G_{22} = -0.5$ (ends at $x = 316.7$), and $G_{32} = -0.125$ (ends at $x = 316.7$). *(our reading of the glyph positions)*

The measurement is stored in `simulation/out/qtl1/glyphs.json`, committed in `b4e8a3c594f12f419560d5cb2a569b649ae93a55`. The command that re-reads the same page and checks those coordinates is

```
python3formalization/correlated-residuals/simulation/measure_glyphs.py--
pdflocal-extraction/literature/unl-sem/mi-qtl2010.pdf--checkformalizatio
n/correlated-residuals/simulation/out/qtl1/glyphs.json
```

The PDF is not in the repository. The copy used here has SHA-256 `27c4f3926c9b5af9b399c9b3ef36918c28e7967ef777444b620279dd91628a81`.

3.2 Table 2, including 0.75

With $B = M$ and the signed G from the glyph reading, $(I - B)^{-1}G$ has $(1, 3)$ entry $3/4$. The rest of the parameter block matches Table 2 except the two QTL2 cells discussed in Section 4.4. The command is

```
python3formalization/correlated-residuals/simulation/table2_totals.py
```

It writes `simulation/out/table2_totals.json` and exits with status 1 if the $(1, 3)$ entry is not $3/4$ or the QTL2 middle total is not $-7/16$. *(our own calculation)*

The earlier reading $B = -M$ with unsigned G gives $1/4$ in that cell. An intermediate reading, $B = +M$ with unsigned G, gives $7/8$. Neither is 0.75.

3.3 The single-trait power on the earlier reading

The 13% figure is the single-trait detection rate for QTL3 on Y_1 under $B = -M$ and unsigned G, 100 replicates, seed `20101027`. It is recomputed by `simulation/gaps.py`, which writes `simulation/out/gaps.json` and `simulation/out/gaps_table.csv`, committed in `c2769042e51806a77b2b70237fb4c014552e78fb`.

```
cd formalization/correlated-residuals/simulation
python3 gaps.py
```

Under the glyph reading the same cell is 86.2% at 1000 replicates, seed `20101041`, in that file. The printed value is 97. The glyph reading accounts for most of the difference between 13 and 97. It does not meet 97. *(our own simulation)*

3.4 Where the 2008 counts sit

The 2008 conference paper, DOI `10.4148/2475-7772.1095`, SHA-256 `ab2cc7ddc94025609cdaa83a6353b22dcdbb4693fedf2b70df71f5fc8aa1452a`, is a different design: 300 lines and one QTL at 42 cM. Its Table 2 text layer gives multi-trait SEM overall power 91, multi-trait least-squares overall power 90, and single-trait powers 2, 39, 4, and overall 42. Trait-specific cells of the two multi-trait rows were not in the text layer. Those counts are not a relabelling of SAGMB Table 1. The file is recorded in `SOURCES.md`. Our three-QTL scan at 300 lines, which is not the conference design, gave SEM powers 89.3, 85.0, and 99.9. *(taken from a cited source, for the conference counts)*

4 What remains unresolved

The language in this section is graded on purpose. A difference between a printed number and a number computed here is a source discrepancy under our reading, or a gap this search did not close. It is not, by itself, evidence that the printed number is a mistake.

4.1 The gap between 64/77/96 and 51/54/86

Table 1 prints SEM powers 64, 77, and 96 for the three QTLs. The glyph baseline at the printed positions, $n = 150$, threshold 14.86, and 1000 replicates gives 50.8, 53.5, and 86.1 (`simulation/out/assumptions/baseline.json`, seed `20101101`, commit `82ff59d6bcfb02c273aaf1f83195e0d120fe0915`). Rounded to the nearest percent, the comparison is 64/77/96 against 51/54/86. The command is

```
cd formalization/correlated-residuals/simulation
python3 assumptions.py --configs baseline --reps 1000
```

The absolute gaps are about 13, 24, and 10 points. QTL2 is the large one. A second seed stays in the same range. Runs that change one modeling choice at a time are in Section 12. None of the choices the simulation paragraph leaves open closes the QTL2 gap while keeping the other two cells near the printed triple. *(our own simulation, set next to a cited table)*

Two calculations reproduce the middle cell and miss the first. At $n = 250$ and the printed positions the SEM powers are 81.6, 77.1, and 98.7. The analytic approximation with the other QTL genotypes controlled, at $n = 150$, is about 81, 77, and 100. Both put QTL1 near 80 rather than near 64. Replacing Ψ by I gives about 84, 80, and 99, which is the same shape and contradicts equation (8). An oracle scan that uses the realized genotype gives about 65, 64, and 94. A χ^2

threshold on 3 degrees of freedom, which is not the printed 14.86 on 4 degrees of freedom, gives 62.6, 64.6, and 92.9 on the baseline.

4.2 No text-consistent design matched all three numbers

`qtl1_search.py` varies the first QTL position, the sample size, a marker gap around QTL1, local distortion, residual heterozygosity, a two-plot line mean, and two cofactor rules that the simulation sentence does not describe. Each specification is tagged with whether it is consistent with the printed sentence. The search did not find a specification that is consistent with that sentence and close to all three of 64, 77, and 96.

The smallest sum of squared errors among the percentage triples computed in that search is the inconsistent point QTL1 at 24 cM and $n = 250$: SEM powers 65.2, 73.3, and 99.3, squared error about 26, of which the third cell is the larger share. The output is `simulation/out/qtl1/qtl1_24_n250.json`, commit `b4e8a3c594f12f419560d5cb2a569b649ae93a55`.

```
cd formalization/correlated-residuals/simulation
python3 qtl1_search.py --configs qtl1_24_n250 --reps 1000
```

Moving QTL1 off 22 cM pulls the first power toward 64 and pulls the second down from 77. The 99.3 against a printed 96, read as a Binomial$(100, 0.993)$ count, has lower-tail probability about 0.006 for a count of 96 or fewer. That comparison is descriptive. It does not identify a mechanism.

A global-maximum reading, one peak on the chromosome, produces powers on the order of 16, 14, and 65. The printed powers sum to 237, so a rule that returns only one detection cannot be the source of all three printed numbers. Reading 14.86 as a LOD threshold produces power 0 at these effect sizes.

4.3 A probability on the order of 10^{-12}

Table 1 is a 100-replicate experiment. Taking the baseline rates 0.508, 0.535, and 0.861 as the success probabilities,

```
python3formalization/correlated-residuals/simulation/printed_probability
.py
```

writes `simulation/out/printed_probability.json`. The output is the product of three binomial probability mass functions, $\mathrm{Bin}(100, p_i)$ at 64, 77, and 96. The product is 1.37×10^{-12}.

The product treats the three QTL counts as independent. They are computed from the same replicates, so they are not independent. The figure is an order of magnitude under that extra assumption. It says that a 100-replicate draw from the baseline rates is a poor match to the printed triple. It does not say how the printed triple was obtained. *(our own calculation, under an independence assumption we flag)*

4.4 The printed pair -0.435 and 0.065

Table 2, and Table 3 of the Genetics Research companion (the ARS file Man851), print -0.435 and 0.065 for the QTL2 effects that the glyph arithmetic gives as $-7/16 = -0.4375$ and $1/16 = 0.0625$. The same two decimals appear in both papers we could read. Rounding -0.4375 to three decimals is -0.438, not -0.435, and rounding 0.0625 to three decimals is 0.063 or 0.062 according to the tie rule, not 0.065.

This is a source discrepancy under our reading of which cells those printed numbers occupy. It is consistent with a rounding of a different intermediate quantity, with a typesetting substitution, or with a coefficient we have not reconstructed. The two papers agreeing with each other makes a one-time typographical slip a weaker candidate, and it does not choose among the others. *(source discrepancy (our reading))*

4.5 SEM power above least-squares power

On the recursive model the multi-trait SEM likelihood ratio and the multi-trait least-squares likelihood ratio are the same statistic. In the baseline profiles the maximum absolute difference of the mean profiles is about 7×10^{-14}. Table 1 nevertheless prints SEM powers 7, 8, and 2 points above the least-squares powers (64 against 57, 77 against 69, 96 against 94). Figure 2 does not print a numeric scale that would let those gaps be read off the plot.

Equal likelihood ratios make equal powers under a common threshold. A difference of that size, repeated at all three QTLs, is a source discrepancy under our reading of the two columns. It would also arise if the two columns were computed under different thresholds or different cofactor sets. The simulation paragraph describes one scan. The present runs do not separate those possibilities. *(source discrepancy (our reading))*

5 Code and source availability

Each item was checked. The result is the result of that check.

- The 2008 conference paper (DOI `10.4148/2475-7772.1095`) contains no code and no code supplement. The software sentence names SAS or Mplus. SHA-256 of the PDF we read: `ab2cc7ddc94025609cdaa83a6353b22dcdbb4693fedf2b70df71f5fc8aa1452a`.

- The De Gruyter page for SAGMB article 38 (DOI `10.2202/1544-6115.1552`) showed no supplement, and the version of record is paywalled. The file replicated here is the ARS manuscript Man850, SHA-256 `27c4f3926c9b5af9b399c9b3ef36918c28e7967ef777444b620279dd91628a81`. The publisher file was not compared with that manuscript. The request for the version of record is https://github.com/alectionary/vibe-math/issues/19.

- Mi's dissertation, *Genetic applications using structural equation modeling*, University of Nebraska–Lincoln, 2009, is restricted. The catalog page is https://digitalcommons.unl.edu/dissertations/AAI3350451/ and the ProQuest record is document `304940336`. The full text was not obtained. The acquisition request is https://github.com/alectionary/vibe-math/issues/18.

- The Genetics Research companion (ARS file Man851, SHA-256 `630a5e7b90665fd680d58915f6f9fe45e5051e8a5f6fff6af2cf16f8ae3b3739`) says that the programs were written in SAS PROC IML and are available by email to `xjmixu@yahoo.com`. No message was sent. The request for those programs, without sending the email, is https://github.com/alectionary/vibe-math/issues/20.

- Xiaojuan Mi works in industry as a contractor for Eli Lilly via Tech Data Services, and was formerly at Duke Clinical Research Institute. No academic webpage was found. Kent Eskridge's faculty page was not used as a substitute for the code or for the dissertation.

6 Next steps

Three steps are open, and this note does not carry them out.

The dissertation is the document most likely to contain the SAS code, the marker list, and any appendix the articles omit. The route is an interlibrary loan or another institutional copy. Filing that request is outside this replication. The queue entry is issue 18, cited above.

A letter to Kent Eskridge, and a letter to Xiaojuan Mi, could ask for the PROC IML programs and for the design details that decide among the readings in Section 4. Those letters are to be drafted and sent by a person. This replication does not send email and does not write the letters.

The formalization that this repository can continue without those documents is local. The wheat fit is still blocked on the line identifiers, the ANOVA residuals, the ± 1 codes, and the edge set of Figure 5. The Drton, Foygel, and Sullivant characterization of generically identifiable mixed graphs, the half-trek gap of Foygel, Draisma, and Drton, and a polynomial-time ancestral decomposition in the sense of Drton and Weihs are recorded as separate questions (L-F145, L-F146, L-F147) and are not theorems here. A Lean proof of the delta-method central limit theorem is the natural next statement about the plug-in that is already proved consistent.

7 The question, as they pose it

The source is Mi et al., Statistical Applications in Genetics and Molecular Biology 9 (2010), article 38. On printed page 18 (PDF page 20) they write:

> Here, we assumed that the residual errors of the traits were independent which we believe is a reasonable assumption since we removed the effects of environment by using, for the y values in equation (2), the residuals from a main effects linear model with environment and block within environment as the main effects. However, even after removing environment main effects, the traits may still be correlated and the trait error terms can be modeled as being correlated with each other.

(taken from a cited source) The parent request is that extension. It is not a numbered conjecture. Mi's 2009 dissertation (ProQuest AAI3350451) is the obvious place a fuller design might have been written. It is restricted to a ProQuest preview. It was not consulted.

8 The restriction

Equation (7) is the only fully printed structural pattern. On PDF page 10 the equals sign ends at $x = 145.9$ points and the parenthesis of M begins at $x = 148.1$. No operator fits between them, so the display is $y = My + Gx + \varepsilon$. The minus at $x = 189.1$ is inside M, on the middle row, and three further minuses sit on G_{11}, G_{22}, and G_{32}. *(our reading of the glyph positions)*

$$M = \begin{pmatrix} 0 & 1/2 & 1/4 \\ 0 & 0 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1/8 & 1/4 & 1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

Equation (8) is $\Psi = \mathrm{diag}(1.6, 1.8, 2.5)$. *(taken from a cited source)* In $y = By + Gx + \varepsilon$ one then has $B = M$. A first reading placed the interior minus in front of M and took every entry of G as positive. That reading is the population of the first replication below. The zero pattern is the same either way: the free positions are still $(1, 2)$, $(1, 3)$, and $(2, 3)$. The wheat Figure 5 graph has four traits and is not this matrix. The restriction is covariance identification for this three-trait recursive

pattern: diagonal Ψ, one extra residual covariance, and the three ways to drop one edge and free the residual pair that edge had used. Singular Ψ is outside the claim. The general mixed-graph characterization is not this restriction.

9 What we proved

The following are Lean proofs on the positive-definite domain. The axioms are `propext`, `Classical.choice`, and `Quot.sound`. There is no `sorry`. *(our own Lean proof)*

- `recover_sigma`: diagonal residuals and positive-definite Φ are a left inverse of the joint covariance on the equation-(7) support.
- `bow12_same_law` and its two siblings: each one-edge residual extension collides, for every exogenous Γ.
- `recover_bow`, `recover_bow01`, `recover_bow23`: each bow-free missing-edge pattern is a left inverse.
- `unitUpper_factor_unique` and `sigmaRec_injective`: in every trait and covariate dimension, a recursive pattern with positive diagonal residuals has a unique unitriangular factorization, and the joint blocks determine (B, Ψ, Γ, Φ).
- `plug_in_consistent_ae` and the three bow-free analogues `plug_in_bow_consistent_ae`, `plug_in_bow01_consistent_ae`, `plug_in_bow23_consistent_ae`: pairwise i.i.d. samples with integrable coordinate products make those plug-ins almost surely consistent.

The Gaussian delta-method formula for the diagonal plug-in is a written derivative together with Isserlis' covariance, checked numerically in `delta_method.py`. *(our own written proof, and our own simulation)* It is not a Lean central limit theorem, and it is not listed among the proved results.

Statement	Lean declaration	File and lines
Diagonal left inverse	`recover_sigma`	`Diagonal.lean`, 246–279
One-edge collision	`bow12_same_law` and two siblings	`Collision.lean`, 114–160
Bow-free left inverse	`recover_bow`	`BowFree.lean`, 187–213
Other missing edges	`recover_bow01`, `recover_bow23`	`BowMissing.lean`, 148 and 353
Triangular factor	`unitUpper_factor_unique`	`Triangular.lean`, 68–135
Recursive identification	`sigmaRec_injective`	`Triangular.lean`, 298–317
Almost-sure consistency	`plug_in_consistent_ae` and `plug_in_bow_consistent_ae`	`Law.lean`, 174–199; `BowConsistency.lean`, 270–304

The links are pinned to commit `05d69ea141adf3d5ee936d3b48061bde241250dc`, which contains these Lean files. The simulation script and the JSON outputs are commit `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`. Appendix A prints the central proof bodies.

Brito and Pearl (Structural Equation Modeling, 2002) prove that a recursive linear model is identifiable when no residual covariance joins a variable to its direct effect. Our three bow-free left inverses are the three-trait case of that pattern, proved from the printed matrices rather than cited as a substitute for the algebra. Drton, Foygel, and Sullivant (Annals of Statistics, 2011) give a global criterion for mixed graphs; the half-trek criterion of Foygel, Draisma, and Drton (Annals of Statistics, 2012) is a generic sufficient condition. Neither is re-proved here, and the 2011 characterization stays a note with no inventory id. Li et al. (PLoS Genetics, 2006) and Valente, Rosa, de los Campos, Gianola, and Silva (Genetics, 2010) are earlier structural and recursive

treatments of multiple traits. Hidalgo-Contreras, Salinas-Ruiz, Eskridge, and Baenziger (Agronomy, 2021) fit a later Nebraska wheat SEM; their lines are a 2013 F6 panel, not the 1999–2001 RICLs, and none of their estimates enter the simulations.

10 Replication design

The printed design, PDF pages 9–10, is 100 replicates of 150 RIL lines, one chromosome of 100 cM, 11 evenly spaced markers, additive QTLs at 22, 42, and 78 cM, phenotypes from equations (7)–(8), and a test every 1 cM. QTL genotypes are coded $+1$ and -1. Cofactors are all markers except the flanking markers. The printed critical values are 14.86 and 10.60. *(taken from a cited source)* Those two numbers are the chi-square 0.005 quantiles on 4 and 2 degrees of freedom (14.860 and 10.597). The paragraph on PDF page 9 writes the Bonferroni approximation $\chi^2_{\alpha/M,\,m+1}$. Ten marker intervals and $\alpha = 0.05$ give the tail probability 0.005, with $m + 1 = 4$ for three traits and $m + 1 = 2$ for one trait. *(our own calculation, from the cited formula)*

The paper does not state the following, so each is our assumption. *(our assumption)*

1. The map function is Haldane, with no interference.
2. An RIL is an F1 selfed for 12 generations. A leftover heterozygote is coded ± 1 with equal probability. There is no segregation distortion.
3. The eleven markers are at $0, 10, \ldots, 100$ cM.
4. The analysis uses the centered covariance with divisor n, which is the fitting function (6). We did not run SAS PROC TCALIS.
5. The flanking expectation is the conditional mean of the QTL genotype under that same 12-generation law. The 10 cM window mentioned later in the paper is for the wheat scan, not for this simulation, and was not used.
6. A named QTL is detected when the maximum likelihood ratio in $[0, 32]$, $(32, 60]$, or $(60, 100]$ exceeds the threshold. Effects are read at that argmax.

Seeds are `20101027` for this study. The outputs are `study1.json`, `study1_power.csv`, and `study1_effects.csv` under `simulation/out`.

For a complete recursive order, the Gaussian likelihood of the diagonal structural model equals the likelihood of the multivariate regression on the same exogenous regressors: the sum of the successive residual log variances is the log determinant of the residual covariance. The two joint likelihood ratios are therefore the same number. In the run, the maximum absolute difference of the two mean profiles is 7×10^{-14}. *(our own written proof, confirmed by our own simulation)* Mi et al. describe the two profiles as very close (PDF page 14). Their Table 1 powers still differ by 7, 8, and 2 replicates out of 100.

11 Replication results

Table 1 compares detection percentages. Our multi-trait powers keep the printed order, third QTL above the first two, and they sit 5 to 16 points below Table 1. The single-trait column for the first trait at the first QTL matches Table 1 exactly (6%). The single-trait column for the first trait at the third QTL does not: Table 1 prints 97% and the replication gives 13%.

Table 1: Detection percentages. "Mi" is Table 1 on PDF page 12. "Ours" is 100 replicates, seed `20101027`. SEM and LS coincide in our run.

QTL	SEM Mi	SEM ours	LS Mi	ST Y_1 Mi	ST Y_1 ours	ST overall ours
1	64	59	57	6	6	24
2	77	61	69	3	0	35
3	96	86	94	97	13	34

The algebraic total effects under $B = -M$ are, by columns of G,

$$(I + M)^{-1}G = \begin{pmatrix} -1/8 & 0 & 1/4 \\ 3/8 & 7/16 & 3/8 \\ 1/4 & 1/8 & 1/4 \end{pmatrix}.$$

(our own calculation) Table 2's parameter block matches this matrix on Y_2 and Y_3 for the first and third QTLs, and it does not match it on Y_1. The printed total effect of the third QTL on Y_1 is 0.75, against 0.25 from this first reading. Section 12 tests that disagreement directly.

Direct-effect means at the SEM argmax are close to G where the signal is large. For the third QTL they are 0.460 (0.175), 0.476 (0.179), and 0.242 (0.190), against the printed SEM estimates 0.479 (0.176), 0.527 (0.166), and 0.257 (0.205) and against $G = (1/2, 1/2, 1/4)$. *(our own simulation, set next to a cited source)* Position means are 22.65 (7.28), 41.26 (5.25), and 79.85 (3.76), against the printed 21.57 (2.83), 41.95 (2.58), and 78.03 (2.16). The biases are small. Our standard deviations are larger. The mean profile exceeds 14.86 only at 40 cM and on 75–82 cM. Figure 2 is described as one simulated population, with excursions on 16–28, 38–48, and 70–88 cM. An average profile is a different object, and ours is lower.

12 Which hypothesis closes which gap

Five changes were rerun from fixed seeds. The full grid is `simulation/out/gaps.json` and `gaps_table.csv`. Closure is $1 - |\hat{p} - p_{\text{printed}}|/|p_{\text{first}} - p_{\text{printed}}|$, using the first reading's 100-replicate powers as the denominator. A value of 1 means the printed number is met. A negative value means the estimate moved away from the table. *(our own simulation)*

Table 2: Gap closure. SEM powers are the three QTLs. ST_3 is single-trait power for QTL3 on Y_1 (printed 97). ST_2 is single-trait power for QTL2 on Y_1 (printed 3). SDs are the three SEM position standard deviations. T_{31} is the algebraic total effect of QTL3 on Y_1 (printed 0.75).

Reading	SEM	ST_3	ST_2	position SD	T_{31}	ST_3 closed	T_{31} closed
Printed	64, 77, 96	97	3	2.83, 2.58, 2.16	0.75	—	—
First, $n = 100$	59, 61, 86	13	0	7.28, 5.25, 3.76	0.25	0	0
First, $n = 1000$	65, 63, 87	9	1	6.20, 5.32, 4.38	0.25	−0.05	0
$B = +M$, G unsigned	68, 57, 86	87	44	5.90, 5.33, 4.60	0.875	0.88	0.75
Glyph reading, $n = 100$	46, 45, 80	85	1	5.57, 5.93, 5.10	0.75	0.86	1
Glyph, $n = 1000$	52, 48, 87	86	1	5.15, 5.42, 3.47	0.75	0.87	1
Glyph, no cofactors	46, 65, 100	100	65	6.99, 9.12, 3.80	0.75	0.96	1
Glyph, wheat cofactors	34, 24, 89	95	1	7.33, 7.91, 3.77	0.75	0.98	1
Glyph, $\Psi = I$	80, 78, 98	97	2	4.86, 4.02, 2.79	0.75	1.00	1
Glyph, window ±5 cM	43, 41, 80	83	1	2.80, 2.76, 2.03	0.75	0.83	1
Glyph, $\alpha = 0.05$	77, 84, 98	95	14	5.57, 5.93, 5.10	0.75	0.98	1

Sign convention. $B = +M$ with unsigned G gives total effect 0.875 on Y_1 for QTL3, not 0.75.

The glyph reading gives

$$(I - M)^{-1}G = \begin{pmatrix} 1/8 & 0 & 3/4 \\ 3/8 & -7/16 & 3/8 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

(our own calculation) Every Table 2 parameter cell matches this arithmetic except the QTL2 row for Y_2, where the exact values are $-7/16 = -0.4375$ and $1/16 = 0.0625$ and the table prints -0.435 and 0.065. *(source discrepancy (our reading))* The first reading's total of 0.25 is what produced the single-trait power 13%. Under the glyph reading, 1000 replicates (seed `20101041`) give 86.2% (Clopper–Pearson interval 83.9–88.3). That closes most of the 97 versus 13 gap and does not meet 97.

Scan method. The simulation paragraph on PDF page 10 fits every marker except the flanking pair, and the methods section replaces the QTL genotype by its flanking-marker expectation (Haley and Knott, 1992). Dropping the cofactors, on the glyph reading, raises the single-trait QTL3 power to 100% and the SEM power for QTL3 to 100%, and it leaves the first SEM power at 46. The wheat rule on PDF page 11, a $P < 0.1$ screen with at most five cofactors and a 10 cM window, gives single-trait power 95% (interval 88.7–98.4, which contains 97) and SEM powers 34, 24, and 89. Nearest-marker regression gives the same pattern as dropping cofactors. None of these is the sentence printed for the simulation, and none of them brings all three SEM powers onto 64, 77, and 96.

Threshold and window. One hundred phenotype shuffles of a glyph-reading replicate have 95% quantile 14.74 for the maximum likelihood ratio, next to the printed 14.86. A permutation threshold at $\alpha = 0.05$ does not move the powers. The 500 permutations in the paper are in the wheat section. Replacing 14.86 and 10.60 by the $\alpha = 0.05$ quantiles on 4 and 2 degrees of freedom (9.49 and 5.99) raises the glyph-reading SEM powers to 77, 84, and 98 and the single-trait QTL3 power to 95. That threshold is not the printed one. Taking the peak within 5 cM of the true site, under the glyph reading, gives position standard deviations 2.80, 2.76, and 2.03, against 2.83, 2.58, and 2.16. The midpoint windows used in the first replication were our assumption. The paper does not name the window that assigns a peak to a named QTL.

Unprinted design. Eleven evenly spaced markers on a segment of 100 cM, with markers at the ends, are at $0, 10, \ldots, 100$. Shifting them to $5, 15, \ldots, 95$ does not close the SEM gap. Standardizing each trait to sample standard deviation 1 leaves the likelihood ratio essentially unchanged. Replacing Ψ by I does close the single-trait QTL3 gap (97%) and puts the second and third SEM powers on 78 and 98, while the first SEM power rises to 80 (interval 70.8–87.3, which does not contain 64). The simulation sentence says the residuals are drawn from matrix (8). $\Psi = I$ contradicts that sentence. It is a sensitivity, not the display.

Monte Carlo error. At 1000 replicates the first reading gives SEM powers 65.1, 62.5, and 86.6, with intervals 62.1–68.1, 59.4–65.5, and 84.3–88.7. The printed 64 lies in the first interval. The printed 77 and 96 do not. Single-trait power for QTL2 on Y_1 is 1.2% (interval 0.6–2.1) under that reading and 0.8% (0.3–1.6) under the glyph reading. The printed 3 is inside the $n = 100$ interval for zero detections (0–3.6) and outside the $n = 1000$ intervals. The discrepancy is three replicates in Table 1.

The most likely account is the glyph reading of equation (7), together with a peak window of about 5 cM that the paper does not state. That reading removes the 0.75 discrepancy and most of the single-trait QTL3 discrepancy, and the narrow window removes the position-SD discrepancy. It does not remove the SEM gap for QTL2 and QTL3. Those two percentages move onto the printed values only in runs that drop the printed cofactors or replace matrix (8). Both of those runs conflict with a sentence on PDF page 10.

13 Explanations considered

The gaps that remained after the glyph reading, on a fresh 1000-replicate run (seed `20101101`), are SEM powers 50.8, 53.5 and 86.1 against 64, 77 and 96, single-trait power for QTL3 on Y_1 of 88.6 against 97, and single-trait power for QTL2 on Y_1 of 1.2 against 3. The intervals are 50.4–56.6, 83.8–88.2, 86.5–90.5 and 0.6–2.1. A second seed (`20101102`) gives SEM powers 49.4, 48.3 and 84.6. The QTL2 gap of about 24 points is not a seed effect. Family A changes one modeling choice at a time. Family B compares the printed counts with the arithmetic, with a noncentral chi-square approximation, and with the companion designs. The grid is `simulation/out/assumptions/`. *(our own simulation)*

Table 3: Explanations considered. Closure is the fraction of the seed-`20101101` gap removed. The four fractions are QTL2 SEM, QTL3 SEM, single-trait QTL3 on Y_1, and single-trait QTL2 on Y_1.

Hypothesis	Family	Evidence	Gap closed	Verdict
Infinite RIL, $2r/(1+2r)$	A	SEM 56, 55, 91	.07/.54/.36/ − .06	open, and too small
Kosambi, infinite RIL	A	SEM 56, 54, 91	.03/.49/.31/ − .06	open in SAGMB; not Table 1
Flanking formula	A	within the seed range	about 0	cannot be distinguished from the 12-generation expectation
Per-equation tests	A	any equation above 10.60 gives 52, 52, 78	negative on QTL3	does not match
Permutation threshold	A	95% quantile 16.1 in 200 shuffles	power falls	14.86 is slightly liberal; the gap is not a low threshold
$\Psi = I$	A	SEM 84, 80, 99; $ST_3 = 98$	.89/.70/.87/.06	contradicts matrix (8)
Entries used as SDs	A	power falls to 24, 25, 59	negative	not the covariance sentence
Half contrast, 0/1	A	power falls to 11, 9, 23	negative	the sentence says +1 and −1
F2 or backcross	A	power falls	negative	the sentence says RIL
Oracle genotype	A	SEM 65, 64, 94; $ST_3 = 95$	.46/.85/.80/.94	not the flanking expectation
QTLs on markers	A	SEM 72, 73, 96	.84/.97/.85/.22	contradicts 22, 42, 78
No cofactors	A	ST_2 rises to 66	.59/.65/.70 and ST_2 breaks	contradicts the cofactor sentence
$n = 250$	B	SEM 82, 77, 99; Y_2 powers 31, 38, 32	1.00/.73/.76/.56	matches QTL2, not QTL1
$n = 300$	B	SEM 89, 85, 100	overshoots QTL2	above Table 1
Chi-square, QTLs controlled	B	powers 81, 77, 100 at $n = 150$	QTL2 matches	not the marker-cofactor scan

No choice the simulation paragraph leaves open closes the QTL2 gap. The infinite-RIL formula is the one printed in the 2008 conference note, and Kosambi is unnamed in SAGMB; together they move QTL2 from 53.5 to 54.2. The no-double-crossover expectation in that note, and an F2 expectation applied to RIL data, stay inside the seed range of the baseline. A 95% permutation threshold is 16.1, and the median of 40 per-replicate thresholds, each the maximum of 19 null scans, is 16.9. Both are above 14.86, so a permutation rule makes the powers smaller (43.5, 44.2, 82.0 at 16.1). Replacing 14.86 by the $\alpha = 0.05$ quantile 9.49 raises the baseline to 84.7, 82.6 and 98.5, which is not the printed critical value. Standardizing the traits, scanning every 2 or 5 cM, and shifting the eleven markers off the ends do not close QTL2. The ±1 and 0/2 codes have the same homozygote contrast, so they have the same likelihood ratio; a 0/1 code with the printed G halves that contrast and the power collapses. *(our own simulation)*

The printed 77 is consistent with two calculations that are not the printed marker scan. With the three QTL genotypes in each structural equation, the noncentral chi-square approximation at $n = 150$ and threshold 14.86 gives joint powers 81%, 77% and 100%. The partial single-trait power for QTL3 on Y_1 is 99.7%, and for QTL2 on Y_1 it is 0.1%. The same approximation at $n = 250$ and $n = 300$ is above 97% for every QTL. *(our own calculation)* The marker-cofactor scan at 250 lines, the sample size of Mi et al. Genetics Research (2010), gives SEM powers 81.6, 77.1 and 98.7, single-trait Y_1 powers 3.2, 2.2 and 99.0, and single-trait Y_2 powers 31, 38 and 32 against the printed 29, 43 and 33. QTL2's SEM cell matches. QTL1's SEM cell is 82 against 64. At 300 lines, the sample size of the 2008 conference note, the three-QTL scan is 89.3, 85.0 and 99.9, above Table 1. *(our own simulation)*

Those companion designs are not Table 1. Genetics Research simulates 250 lines and two QTLs, at 42 and 78 cM, with codes 2 and 0, and the detection rule is a posterior peak rather than the likelihood-ratio threshold. Its Table 2 powers are 99 and 100. Equation (15) and Figure 2 of that paper print the same path signs as the glyph reading, 1/2, 1/4 and $-1/2$, and the same rounded QTL2 row -0.435 and 0.065. *(taken from a cited source)* The conference note simulates 300 lines and one QTL at 42 cM, names Haldane's map and the expansion $2r/(1+2r)$, and uses codes 2 and 0. Its Table 2 text layer gives multi-trait SEM overall power 91, multi-trait least-squares overall power 90, and single-trait powers 2, 39, 4, and overall 42. Trait-specific cells of the two multi-trait rows were not in the text layer. Neither table is a relabeling of SAGMB Table 1. *(taken from a cited source)*

Table 2's QTL2 row prints -0.435 and 0.065 where the path product is $-7/16 = -0.4375$ and $1/16 = 0.0625$. The same printed decimals appear in the Genetics Research table. *(source discrepancy (our reading))* On a complete recursive model the multi-trait SEM and multi-trait LS likelihood ratios are the same number, and in every run here their powers agree. Table 1 prints SEM powers 7, 8 and 2 points above the LS powers, while the text says the two profiles are very close. *(source discrepancy (our reading))* Figure 2 is described as one simulated population and does not print a numeric likelihood-ratio scale, so the power counts cannot be read off the figure.

The best account is that no open choice in the simulation paragraph produces Table 1, and that the cell 77 is consistent both with the chi-square power of these direct effects at 150 lines when the other QTL genotypes are controlled and with the marker scan at 250 lines. Both of those readings put the first QTL near 80% rather than at the printed 64%. The single-trait QTL2 cell, 3 against our 1.2, is consistent with the algebraic total effect of zero.

13.1 A joint reading of 64, 77 and 96

A second block asked which reading could produce the SEM triple 64, 77, 96 together. Every entry of equation (7) was measured again on PDF page 10. The minus signs are four glyphs. One ends at $x = 250.7$ on $G_{11} = -1/8$. One ends at $x = 194.5$ on $M_{23} = -1/2$. Two end at $x = 316.7$, on $G_{22} = -1/2$ and $G_{32} = -1/8$. No other entry carries a minus, and there is no minus between the equals sign and M. The Table 1 digits are the characters 6 and 4, then 77, then 96. *(taken from a cited source)* The scans below use 1000 replicates. Squared error is the sum of squared differences of the three SEM percentages. The chi-square treats each printed percentage as a count out of 100 draws from the simulated rate. *(our own simulation)*

The smallest squared error is 65.2, 73.3, 99.3. Single-trait powers on Y_1 are 2.6, 1.9, 99.1 against 6, 3, 97, and on Y_2 they are 22, 37, 33 against 29, 43, 33. QTL1's 65 cannot be distinguished from the printed 64 at the precision of 100 replicates. QTL2 remains about four points low. QTL3 at 99.3 makes a printed 96 unlikely if that rate were true: the lower-tail probability is about 0.006. The position 24 cM and the sample size 250 are not the simulation sentence. Moving QTL1 from

Table 4: SEM triples ranked by squared error against 64, 77, 96. The chi-square is for those three cells. A text-consistent row is one the simulation paragraph on PDF page 10 states.

Hypothesis	Family	SEM powers	SSE	χ^2	Consistent with the text
QTL1 at 24 cM, 250 lines	A+C	65, 73, 99	26	16.4	no
QTL1 at 25 cM, 250 lines	A+C	62, 69, 99	74	13.8	no
QTL1 at 23 cM, 250 lines	A+C	72, 73, 99	80	8.6	no
Oracle genotype, 150 lines	A	65, 64, 94	163	7.4	no
Threshold $\chi^2_{3,0.995}$	B	63, 65, 93	165	8.3	no; printed value is 14.86
Marker gap around QTL1, 250 lines	A+C	78, 76, 98	212	13.6	no
Printed positions, 250 lines	C	82, 77, 99	317	26.3	no; the sentence says 150
Two plots averaged, Ψ at plot level	A	63, 59, 100	341	174	no
Local allele frequency 0.30 at QTL1	A	62, 61, 85	392	20.8	no
$\Psi = I$	A	84, 80, 99	416	39.3	no; contradicts (8)
Glyph baseline, 150 lines	A	51, 54, 86	824	37.4	yes
One peak on the chromosome	B	16, 14, 65	4602	—	the printed sum is 237
14.86 applied to the LOD	B	0, 0, 0	—	—	14.86 is the chi-square value
G transposed	A	55, 94, 19	6308	441	no; columns are labeled QTL

22 to 24 cM is what pulls the first power down from the 82 of the 250-line scan at the printed positions. The same move pulls the second power down from 77 to 73. The two cells do not land on 64 and 77 together. *(our own simulation)*

Family A does not supply a text-consistent joint match. Flipping the sign of G_{11}, which the remeasured glyph does not support, gives 55, 50, 85. Halving G_{21} gives 17, 53, 87. A residual heterozygosity of 0.15 gives 20, 18, 57. Placing QTL1 10 cM from QTL2 gives 24, 22, 84. Kosambi interference was already inside a few points of the baseline. Standardizing the traits was 51, 48, 87.

Family B does not supply the triple either. Stepwise cofactors and a two-stage refit lower the first two powers and raise the third, to about 33, 28, 98 and 29, 28, 98. A marker-bracket window, a 20 cM separation rule, and dropping the terminal 10 cM leave the baseline near 51, 54, 86. The χ^2_3 threshold 12.84 gives 63, 65, 93: the first cell matches, and the second does not. The printed critical value is 14.86, the χ^2_4 quantile. Reading 14.86 as a LOD multiplies the likelihood-ratio threshold by $2 \log 10$ and the powers fall to zero. One peak per chromosome assigns the global maximum to a single QTL, so the three powers sum to at most 100. On the baseline they are 16, 14, 65. The printed powers sum to 237, so that counting rule cannot be the source of all three cells. The flanking regressor has positive variance at every interior test point on this marker grid, so a singular fit does not arise and counting non-convergence as a miss does not change the powers.

Family C. The probability of the printed counts 64, 77, 96 as three independent Bin(100) draws from the baseline rates 0.51, 0.54, 0.86 is about 10^{-12}. Swapping the SEM and LS columns of Table 1 produces 57, 69, 94, which is not a row of any scan here. The PDF prints the digit 6, not 8, so 64 is not an OCR reading of 84. The 250-line scan at the printed positions is 82, 77, 99. That matches the cell 77. It is near 84, 77, 96 in the first cell and not near 64. The Genetics Research powers are 99 and 100 under a different design. The 300-line three-QTL scan is 89, 85, 100.

The best joint account is negative. No tested reading that the simulation paragraph states produces 64, 77, and 96 together. The closest percentage triple uses a QTL position and a sample size that paragraph does not state, and its third cell is still high relative to 96. What remains is that gap: the first cell can be brought near 64 by weakening QTL1 or by a lower threshold, and the second cell can be brought to 77 by using 250 lines or by controlling the other QTL genotypes, and those two adjustments have not been found together in one design the text states.

14 Correlated residuals

The bow drops the edge into trait 0 from trait 2, sets that structural coefficient to zero, and frees the residual covariance $c = \rho\sqrt{1.6 \cdot 2.5}$. The other equation-(7) coefficients stay. The exogenous variables are the three QTL genotypes on the RIL map, not a marker scan. *(our assumption, matching* `recoverBow`*)* The sample is the centered second-moment matrix, $n = 150$, 400 replicates, seed `20101028`.

At the population covariance the bow plug-in recovers the parameter to machine precision. The diagonal plug-in does not. Its probability limit for the missing edge is 0.8ρ: zero at $\rho = 0$, 0.20 at $\rho = 0.25$, and 0.48 at $\rho = 0.6$. The QTL coefficient $G_{11} = 1/8$ has probability limit 0.075 at $\rho = 0.25$ and 0.005 at $\rho = 0.6$. The coefficient $G_{13} = 1/2$ moves to 0.45 and 0.38. *(our own simulation)* A Wald test that the missing edge is zero, using the diagonal delta-method standard error, rejects at rate 0.058 when $\rho = 0$ and at rate 0.728, 0.877, and 1 at $\rho = 0.25$, 0.30, and 0.60. The joint nine-degree Wald test of all QTL coefficients is only mildly inflated at $\rho = 0.25$ (rejection rate 0.065) and reaches 0.297 at $\rho = 0.6$. The damage shows up first as bias in two coefficients and as a spurious edge, not as an immediate explosion of the joint QTL test.

`recoverBow` is centered on the QTL and path coefficients. At $\rho = 0.25$ the Monte Carlo means of b_{12}, b_{23}, G_{11}, and G_{13} are -0.507, -0.498, 0.127, and 0.509. Coverage of the nominal 95% delta-method intervals, with the Gaussian Isserlis covariance plugged into the sample, is between 0.915 and 0.958 across the reported coefficients and the six correlations. The genotypes are ± 1, so Isserlis is an approximation. *(our assumption inside the standard error)* The coverage near 95% says the approximation is tolerable at $n = 150$ for these coefficients. Variance-component means sit a little low (1.54 against 1.6 at $\rho = 0$) because the likelihood uses divisor n; the probability limit is still the population value.

The same seeds were rerun with the glyph-reading coefficients: $b_{12} = +1/2$, $b_{23} = -1/2$, the edge b_{13} removed from $+1/4$, and the signed G. *(our own simulation)* At $\rho = 0.25$ the diagonal probability limit of the missing edge is again 0.20 and the Wald rejection rate is 0.728. `recoverBow` means are $b_{12} = 0.493$, $b_{23} = -0.498$, $G_{11} = -0.123$, and $G_{22} = -0.508$, with coverage between 0.920 and 0.955. The sign reading changes the parameter values and does not change the bias or the coverage.

15 Calibration to three phenotypic correlations

Campbell et al., Crop Science 43 (2003), Table 2, print phenotypic correlations for 95 recombinant inbred chromosome lines pooled across seven environments. There are no standard deviations and no environment or block adjustment. *(taken from a cited source, with the limits just stated)* The text layer of the copy we could read dropped minus signs. The paragraph above the table says that SPSM and kernels per square meter were negatively correlated with thousand-kernel weight. We therefore use $\mathrm{Corr}(\mathrm{TKWT}, \mathrm{SPSM}) = -0.44$. The cells $\mathrm{Corr}(\mathrm{TKWT}, \mathrm{KPS}) = 0.08$ and $\mathrm{Corr}(\mathrm{KPS}, \mathrm{SPSM}) = 0.59$ are the signs in that text layer; the prose does not restate them. *(our assumption for those two signs)* GYLD is the fourth trait in the application and is not in equation (7). The three-trait order is $Y_1 = \mathrm{TKWT}$, $Y_2 = \mathrm{KPS}$, $Y_3 = \mathrm{SPSM}$, which is the chain used in the indirect-effect arithmetic on PDF page 17 of Mi et al. *(our assumption)*

Treating the correlation matrix as a covariance, the diagonal recursive factorization is

$$B = \begin{pmatrix} 0 & 0.521 & -0.747 \\ 0 & 0 & 0.590 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \Psi = \mathrm{diag}(0.629, 0.652, 1).$$

(our own calculation) The eigenvalues of the correlation are 0.224, 1.077, and 1.699, so this sign choice is positive definite. Naive Gaussian standard errors that pretend the matrix is a sample correlation of 95 i.i.d. lines are about 0.10 for the two coefficients in the first equation and 0.084 for 0.590. Campbell pooled seven environments, so those standard errors are not his sampling variances. The point fit does not depend on N.

The scan was rerun with this B and Ψ and with the signed G from the glyph reading of equation (7). Seed `20101029`, 100 replicates. Those QTL coefficients were not estimated from the correlations, and on this residual scale they are large: multi-trait power is 93%, 95%, and 99%. Single-trait power for QTL3 on Y_1 is 96%. That is a consequence of the scale mixture, not evidence that the 2003 correlations produce Mi's powers. *(our own simulation, under an assumption we flag)* The bow check at $\rho = 0.25$ (seed `20101030`) gives a diagonal probability limit 0.198 for the missing edge and a rejection rate 0.74. The unsigned-G run of the same seed had multi-trait powers 100%, 96%, and 100%. The sign change does not turn the calibration into a replication.

16 Limitations

The wheat fit is still blocked. It needs the 98 line identifiers, the ANOVA residuals of grain yield, thousand-kernel weight, kernels per spike, and spikes per square meter, the ± 1 QTL and cofactor codes, the divisor-n sample covariance, and the edge set as a matrix. Figure 5 draws arrows and does not print that matrix. Campbell's Table 2 does not substitute for any of those columns.

The Drton–Foygel–Sullivant criterion is not a theorem of this repository. The delta method is not a Lean central limit theorem. The glyph reading of equation (7) meets Table 2's parameter block up to the two rounded QTL2 entries. It leaves the SEM powers for QTL2 and QTL3, and about eleven points of the single-trait QTL3 power, short of Table 1. A second block tested the choices the simulation paragraph does not pin down. None of them closes the QTL2 gap. A 250-line scan matches that one cell and misses the first QTL. Those leftovers are not Monte Carlo error at 1000 replicates. Twelve generations of selfing, Haldane's map, the marker placement, and the detection windows are our assumptions. The 2021 Hidalgo-Contreras wheat SEM is a different experiment.

17 Reproducibility

The command list for every script is `formalization/correlated-residuals/README.md`. From `formalization/correlated-residuals/simulation`,

```
python3 studies.py --reps 100 --bow-reps 400 --studies 1,2,3
```

writes `study1.json`, `study2.json`, and `study3.json`. The CSV tables are built from those files.

- Simulation commit: `5cd3c9f`.
- Full hash: `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`.
- Branch: `cursor/l-f102-correlated-residuals-4c01`.
- Lean 4.22.0 and Mathlib v4.22.0, built with `lake build`.
- Theorem links are pinned to `05d69ea141adf3d5ee936d3b48061bde241250dc`.
- This PDF is `pdflatex paper.tex`. The ledger is `SOURCES.md`.

- The gap grid is `python3 gaps.py` in the simulation directory. Seeds are `20101027` through `20101042`.

A Central Lean proofs

The diagonal model reads the structural coefficients from the Schur complement $\Omega = \Sigma_{yy} - \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{yx}^{\mathsf{T}}$, then recovers $\Gamma = (I - B)\Sigma_{yx}\Sigma_{xx}^{-1}$ and $\Phi = \Sigma_{xx}$. The bow-free model drops the edge into the first trait from the third and reads the residual covariance c from the same complement. The collision proofs exhibit, for an arbitrary exogenous coefficient, a second parameter point with the same joint blocks. Consistency is continuous mapping of those left inverses, composed with the strong law for the sample Gram matrix. The listings below are the theorem bodies. Their lemmas live in the linked files.

`recover_sigma`, `Diagonal.lean`, lines 246–279.

```
theorem recover_sigma (θ : Params) (h : OnDomain θ) : recover (sigma θ) = θ := by
  obtain ⟨hv1, hv2, hv3, hφ⟩ := h
  have _hpsi : (psi θ).PosDef := by
    rw [psi, posDef_diagonal_iff]
    intro i
    fin_cases i
    · simpa using hv1
    · simpa using hv2
    · simpa using hv3
  have hread : readDiag (omega θ) = (θ.b12, θ.b13, θ.b23, θ.v1, θ.v2, θ.v3) :=
    readDiag_omega θ hv2 hv3
  have hschur := schur_omega θ hφ
  simp only [recover, hschur, hread]
  have hB : (1 -
      ![0, θ.b12, θ.b13;
        0, 0, θ.b23;
        0, 0, 0]) = subB θ := by
    simpa [Bmat] using (subB_eq_one_sub θ).symm
  have hdet : IsUnit θ.phi.det := (isUnit_iff_isUnit_det θ.phi).1 hφ.isUnit
  apply Params.ext
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · simp only [sigma]
    rw [hB]
    calc
      subB θ * (Lmat θ * θ.gamma * θ.phi) * θ.phi⁻¹
          = (subB θ * Lmat θ) * θ.gamma * (θ.phi * θ.phi⁻¹) := by simp only [mul_assoc]
      _ = (1 : M3) * θ.gamma * (1 : M3) := by rw [subB_mul_L, mul_nonsing_inv _ hdet]
      _ = θ.gamma := by simp
  · rfl
```

`recover_bow`, `BowFree.lean`, lines 187–213.

```
theorem recover_bow (θ : Bow) (h : OnBow θ) : recoverBow θ.sigma = θ := by
  obtain ⟨hN, hφ⟩ := h
  have hv2 := Bow.v2_pos θ hN
  have hv3 := Bow.v3_pos θ hN
```

```
  have hread := readBow_omega θ hv2 hv3
  have hschur : θ.sigma.yy - θ.sigma.yx * θ.sigma.xx⁻¹ * θ.sigma.yxᵀ = θ.omega := by
    simpa [Bow.sigma, Bow.omega] using
      schur_noise (L := θ.L) (G := θ.gamma) (Φ := θ.phi) (N := θ.noise) hφ
  simp only [recoverBow, hschur, hread]
  have hB : (1 - !![0, θ.b12, 0; 0, 0, θ.b23; 0, 0, 0]) = θ.subB := by
    simpa [Bow.B] using (Bow.subB_eq θ).symm
  have hdet : IsUnit θ.phi.det := (isUnit_iff_isUnit_det θ.phi).1 hφ.isUnit
  apply Bow.ext
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · simp only [Bow.sigma]
    rw [hB]
    calc
      θ.subB * (θ.L * θ.gamma * θ.phi) * θ.phi⁻¹
          = (θ.subB * θ.L) * θ.gamma * (θ.phi * θ.phi⁻¹) := by simp only [mul_assoc]
      _ = (1 : M3) * θ.gamma * (1 : M3) := by rw [Bow.subB_mul_L, mul_nonsing_inv _ hdet]
      _ = θ.gamma := by simp
  · rfl
```

`bow12_same_law` in `Collision.lean`, lines 114–128. The siblings `bow13_same_law` and `bow23_same_law` repeat the argument on the other two pairs, lines 130–160.

```
theorem bow12_same_law (c : ℝ) (G Φ : M3) :
    bow12L c * (bow12G c * G) * Φ = G * Φ ∧
      bow12L c * ((bow12G c * G) * Φ * (bow12G c * G)ᵀ + bow12G c * (bow12G c)ᵀ) * (bow12L c)ᵀ =
        G * Φ * Gᵀ + 1 := by
  have hG : bow12L c * (bow12G c * G) = (1 : M3) * G := by
    rw [← mul_assoc, bow12_resolve c, one_mul]
  have hN : bow12L c * (bow12G c * (bow12G c)ᵀ) * (bow12L c)ᵀ = (1 : M3) :=
    resolved_noise (bow12_resolve c)
  have hbaseN : (1 : M3) * (1 : M3) * (1 : M3)ᵀ = (1 : M3) := by simp
  obtain ⟨hyx, hyy⟩ := exogenous_lift (L := bow12L c) (G := bow12G c * G)
    (L' := (1 : M3)) (G' := G) (N := bow12G c * (bow12G c)ᵀ) (N' := (1 : M3)) (Φ := Φ)
    hG (hN.trans hbaseN.symm)
  constructor
  · simpa using hyx
  · simpa using hyy
```

`unitUpper_factor_unique`, `Triangular.lean`, lines 68–135.

```
theorem unitUpper_factor_unique {n : ℕ}
    {L M : Matrix (Fin n) (Fin n) ℝ} {d e : Fin n → ℝ}
    (hL : UnitUpper L) (hM : UnitUpper M)
    (hd : ∀ i, 0 < d i) (he : ∀ i, 0 < e i)
    (hΩ : L * diagonal d * Lᵀ = M * diagonal e * Mᵀ) :
    L = M ∧ d = e := by
  induction n with
  | zero =>
    constructor
    · ext i j
      exact i.elim0
    · ext i
      exact i.elim0
```

```
  | succ n ih =>
    have hde_last : d (Fin.last n) = e (Fin.last n) := by
      have hLlast := gram_last (d := d) hL (Fin.last n)
      have hMlast := gram_last (d := e) hM (Fin.last n)
      rw [hL.2] at hLlast
      rw [hM.2] at hMlast
      have hentry := congr_fun (congr_fun hΩ (Fin.last n)) (Fin.last n)
      rw [hLlast, hMlast] at hentry
      simpa using hentry
    have hcol : ∀ i, L i (Fin.last n) = M i (Fin.last n) := by
      intro i
      have hLi := gram_last (d := d) hL i
      have hMi := gram_last (d := e) hM i
      have hentry := congr_fun (congr_fun hΩ i) (Fin.last n)
      rw [hLi, hMi, hde_last] at hentry
      exact mul_right_cancel₀ (he (Fin.last n)).ne' hentry
    have hlead :
        lead L * diagonal (fun k : Fin n => d k.castSucc) * (lead L)ᵀ =
          lead M * diagonal (fun k : Fin n => e k.castSucc) * (lead M)ᵀ := by
      ext i j
      have hLij := lead_gram (L := L) (d := d) i j
      have hMij := lead_gram (L := M) (d := e) i j
      have hentry := congr_fun (congr_fun hΩ i.castSucc) j.castSucc
      rw [hLij, hMij] at hentry
      have hr :
          L i.castSucc (Fin.last n) * d (Fin.last n) * L j.castSucc (Fin.last n) =
            M i.castSucc (Fin.last n) * e (Fin.last n) * M j.castSucc (Fin.last n) := by
        rw [hcol, hcol, hde_last]
      rw [hr] at hentry
      exact add_right_cancel hentry
    have hrec :=
      ih (lead_unitUpper hL) (lead_unitUpper hM) (fun i => hd i.castSucc)
        (fun i => he i.castSucc) hlead
    refine ⟨?_, ?_⟩
    · ext i j
      cases i using Fin.lastCases with
      | last =>
        cases j using Fin.lastCases with
        | last =>
          rw [hL.2, hM.2]
        | cast j =>
          rw [hL.1 _ _ j.castSucc_lt_last, hM.1 _ _ j.castSucc_lt_last]
      | cast i =>
        cases j using Fin.lastCases with
        | last =>
          exact hcol i.castSucc
        | cast j =>
          have hlead_ij := congr_fun (congr_fun hrec.1 i) j
          simpa [lead, submatrix_apply] using hlead_ij
    · ext i
      cases i using Fin.lastCases with
      | last => exact hde_last
      | cast i =>
        have := congr_fun hrec.2 i
        simpa using this
```

sigmaRec_injective, Triangular.lean, lines 298–317.

```
theorem sigmaRec_injective {n p : ℕ} {θ θ' : RecModel n p}
```

```
    (hθ : OnRec θ) (hθ' : OnRec θ') (h : sigmaRec θ = sigmaRec θ') : θ = θ' := by
  have hθ0 := hθ
  have hθ'0 := hθ'
  obtain ⟨hB, hd, hΦ⟩ := hθ
  obtain ⟨hC, he, hΦ'⟩ := hθ'
  have hgram :
      (1 - θ.B)⁻¹ * diagonal θ.d * ((1 - θ.B)⁻¹)ᵀ =
        (1 - θ'.B)⁻¹ * diagonal θ'.d * ((1 - θ'.B)⁻¹)ᵀ := by
    have hθs := schur_sigmaRec θ hΦ
    have hθs' := schur_sigmaRec θ' hΦ'
    rw [h] at hθs
    exact hθs.symm.trans hθs'
  obtain ⟨hBeq, hdeq⟩ := recursive_residual_unique hB hC hd he hgram
  have hγ : θ.gamma = θ'.gamma := by
    have hg := (recover_exogenous θ hθ0).2
    have hg' := (recover_exogenous θ' hθ'0).2
    rw [hg, hg', h, hBeq]
  have hφ : θ.phi = θ'.phi := by simpa [sigmaRec] using congrArg Joint.xx h
  exact RecModel.ext hBeq hdeq hγ hφ
```

plug_in_consistent_ae, Law.lean, lines 174–199.

```
theorem plug_in_consistent_ae {Ω : Type*} [MeasurableSpace Ω] {μ : Measure Ω}
    {Z : ℕ → Ω → Idx → ℝ} {θ : Params} (hθ : OnDomain θ)
    (hindep : Pairwise fun i j => IndepFun (Z i) (Z j) μ)
    (hident : ∀ i, IdentDistrib (Z i) (Z 0) μ μ)
    (hint : ∀ a b : Idx, Integrable (fun ω => Z 0 ω a * Z 0 ω b) μ)
    (hyy : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inl j) ∂μ = (sigma θ).yy i j)
    (hyx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inr j) ∂μ = (sigma θ).yx i j)
    (hxx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inr i) * Z 0 ω (.inr j) ∂μ = (sigma θ).xx i j) :
    ∀ᵐ ω ∂μ, Tendsto (fun n =>
        ((recover (sampleCov Z n ω)).b12, (recover (sampleCov Z n ω)).b13,
          (recover (sampleCov Z n ω)).b23, (recover (sampleCov Z n ω)).v1,
          (recover (sampleCov Z n ω)).v2, (recover (sampleCov Z n ω)).v3,
          (recover (sampleCov Z n ω)).gamma, (recover (sampleCov Z n ω)).phi))
      atTop
      (nhds (θ.b12, θ.b13, θ.b23, θ.v1, θ.v2, θ.v3, θ.gamma, θ.phi)) := by
  filter_upwards [strong_gram Z hindep hident hint] with ω hω
  have hyy' : Tendsto (fun n => (sampleCov Z n ω).yy) atTop (nhds (sigma θ).yy) := by
    simpa [sampleCov, blockYY_pop hyy] using tendsto_blockYY hω
  have hyx' : Tendsto (fun n => (sampleCov Z n ω).yx) atTop (nhds (sigma θ).yx) := by
    simpa [sampleCov, blockYX_pop hyx] using tendsto_blockYX hω
  have hxx' : Tendsto (fun n => (sampleCov Z n ω).xx) atTop (nhds (sigma θ).xx) := by
    simpa [sampleCov, blockXX_pop hxx] using tendsto_blockXX hω
  exact plug_in_consistent hθ hyy' hyx' hxx'
```

plug_in_bow_consistent_ae, BowConsistency.lean, lines 270–304.

```
theorem plug_in_bow_consistent_ae {Ω : Type*} [MeasurableSpace Ω] {μ : Measure Ω}
    {Z : ℕ → Ω → Idx → ℝ} {θ : Bow} (hθ : OnBow θ)
    (hindep : Pairwise fun i j => IndepFun (Z i) (Z j) μ)
    (hident : ∀ i, IdentDistrib (Z i) (Z 0) μ μ)
    (hint : ∀ a b : Idx, Integrable (fun ω => Z 0 ω a * Z 0 ω b) μ)
    (hyy : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inl j) ∂μ = θ.sigma.yy i j)
```

```
  (hyx : ∀ i j : Fin 3,
    ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inr j) ∂μ = θ.sigma.yx i j)
  (hxx : ∀ i j : Fin 3,
    ∫ ω, Z 0 ω (.inr i) * Z 0 ω (.inr j) ∂μ = θ.sigma.xx i j) :
  ∀ᵐ ω ∂μ, Tendsto (fun n =>
      ((recoverBow (sampleCov Z n ω)).b12, (recoverBow (sampleCov Z n ω)).b23,
        (recoverBow (sampleCov Z n ω)).v1, (recoverBow (sampleCov Z n ω)).v2,
        (recoverBow (sampleCov Z n ω)).v3, (recoverBow (sampleCov Z n ω)).c,
        (recoverBow (sampleCov Z n ω)).gamma, (recoverBow (sampleCov Z n ω)).phi))
    atTop
    (nhds (θ.b12, θ.b23, θ.v1, θ.v2, θ.v3, θ.c, θ.gamma, θ.phi)) := by
filter_upwards [strong_gram Z hindep hident hint] with ω hω
have hyy' : Tendsto (fun n => (sampleCov Z n ω).yy) atTop (nhds θ.sigma.yy) := by
  have hpop : blockYY (popGram μ (Z 0)) = θ.sigma.yy := by
    ext i j
    simpa [blockYY, popGram] using hyy i j
  simpa [sampleCov, hpop] using tendsto_blockYY hω
have hyx' : Tendsto (fun n => (sampleCov Z n ω).yx) atTop (nhds θ.sigma.yx) := by
  have hpop : blockYX (popGram μ (Z 0)) = θ.sigma.yx := by
    ext i j
    simpa [blockYX, popGram] using hyx i j
  simpa [sampleCov, hpop] using tendsto_blockYX hω
have hxx' : Tendsto (fun n => (sampleCov Z n ω).xx) atTop (nhds θ.sigma.xx) := by
  have hpop : blockXX (popGram μ (Z 0)) = θ.sigma.xx := by
    ext i j
    simpa [blockXX, popGram] using hxx i j
  simpa [sampleCov, hpop] using tendsto_blockXX hω
exact plug_in_bow_consistent hθ hyy' hyx' hxx'
```

B Artifacts

Each row is one claim in this note, the file that holds it, and the command that regenerates it. Commands are from the repository root unless the row says otherwise. Commits are on `cursor/l-f102-correlated-residuals-4c01`. A short `-reps` run overwrites the committed JSON; the commands below are the runs that match the cited files.

Claim	File and commit	Command
Glyph boxes for equation (7): equals ends at 145.9, parenthesis at 148.1, four unary minuses	`simulation/out/qtl1/glyphs.json`, `b4e8a3c594f12f419560d5cb2a569b649ae93a55`	`python3formalization/correlated-residuals/simulation/measure_glyphs.py--pdflocal-extraction/literature/unl-sem/mi-qtl2010.pdf--checkformalization/correlated-residuals/simulation/out/qtl1/glyphs.json`
Total effects, including 3/4 and $-7/16$	`simulation/out/table2_totals.json`	`python3formalization/correlated-residuals/simulation/table2_totals.py`
Baseline SEM powers 50.8, 53.5, 86.1	`simulation/out/assumptions/baseline.json`, `82ff59d6bcfb02c273aaf1f83195e0d120fe0915`	`python3assumptions.py--configsbaseline--reps1000` in `simulation/`

Claim	File and commit	Command
Gap grid, including the 13% and 86.2% single-trait cells	`simulation/out/gaps.json`, `gaps_table.csv`, `c2769042e51806a77b2b70237fb4c014552e78fb`	`python3gaps.py` in `simulation/`
Closest triple 65.2, 73.3, 99.3	`simulation/out/qtl1/qtl1_24_n250.json`, `b4e8a3c594f12f419560d5cb2a569b649ae93a55`	`python3qtl1_search.py--configsqtl1_24_n250--reps1000` in `simulation/`
Product 1.37×10^{-12}	`simulation/out/printed_probability.json`, from the baseline file above	`python3formalization/correlated-residuals/simulation/printed_probability.py`
Studies 1–3, bow Monte Carlo, Campbell scale check	`simulation/out/study1.json`, `study2.json`, `study3.json`, `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`	`python3studies.py--reps100--bow-reps400--studies1,2,3` in `simulation/`
Identification algebra at the earlier sign convention	`witnesses.json`, written by `check_witness.py`	`python3check_witness.py` in `formalization/correlated-residuals/`
Delta-method numeric check	`delta_check.json`	`python3delta_method.py` in `formalization/correlated-residuals/`
Bow-free plug-in numeric check	`fit_bow.json`	`python3fit_bow.py` in `formalization/correlated-residuals/`
Lean theorems, no `sorry`. Axioms `propext`, `Classical.choice`, `Quot.sound`	`CorrelatedResiduals/*.lean`, listings pinned to `05d69ea141adf3d5ee936d3b48061bde241250dc`	`lakebuild` in `formalization/correlated-residuals/`
This PDF	`paper/paper.pdf`	`pdflatexpaper.tex` in `paper/`

References

[1] C. Brito and J. Pearl. A new identification condition for recursive models with correlated errors. *Structural Equation Modeling* 9 (2002), 459–474.

[2] B. T. Campbell, P. S. Baenziger, K. S. Gill, K. M. Eskridge, H. Budak, M. Erayman, and Y. Yen. Identification of QTLs and environmental interactions associated with agronomic traits on chromosome 3A of wheat. *Crop Science* 43 (2003), 1493–1505.

[3] M. Drton, R. Foygel, and S. Sullivant. Global identifiability of linear structural equation models. *Annals of Statistics* 39 (2011), 865–886.

[4] R. Foygel, J. Draisma, and M. Drton. Half-trek criterion for generic identifiability of linear structural equation models. *Annals of Statistics* 40 (2012), 1682–1713.

[5] J. V. Hidalgo-Contreras, J. Salinas-Ruiz, K. M. Eskridge, and S. P. Baenziger. Incorporating molecular markers and causal structure among traits using a Smith–Hazel index and structural equation models. *Agronomy* 11 (2021), 1953.

[6] R. Li, S.-W. Tsaih, K. Shockley, I. M. Stylianou, J. Wergedal, B. Paigen, and G. A. Churchill. Structural model analysis of multiple quantitative traits. *PLoS Genetics* 2 (2006), e114.

[7] X. Mi, K. Eskridge, and D. Wang. Multi-trait QTL mapping using a structural equation model. Conference on Applied Statistics in Agriculture (2008). DOI `10.4148/2475-7772.1095`.

[8] X. Mi. Genetic applications using structural equation modeling. Doctoral dissertation, University of Nebraska–Lincoln, 2009. ProQuest AAI3350451. Not consulted. The full text was not obtained.

[9] X. Mi, K. Eskridge, D. Wang, P. S. Baenziger, B. T. Campbell, K. S. Gill, I. Dweikat, and J. Bovaird. Regression-based multi-trait QTL mapping using a structural equation model. *Statistical Applications in Genetics and Molecular Biology* 9 (2010), article 38.

[10] X. Mi, K. Eskridge, D. Wang, P. S. Baenziger, B. T. Campbell, K. S. Gill, and I. Dweikat. Bayesian mixture structural equation modelling in multiple-trait QTL mapping. *Genetics Research* 92 (2010), 239–250.

[11] B. D. Valente, G. J. M. Rosa, G. de los Campos, D. Gianola, and M. A. Silva. Searching for Recursive Causal Structures in Multivariate Quantitative Genetics Mixed Models. *Genetics* 185 (2010), 633–644.